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EFFECT OF CONFINEMENT ON THE STEADY FLOW OF BINGHAM PLASTIC FLUIDS PAST A HEATED SQUARE CYLINDER

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ABSTRACT

The Bingham plastic fluids are characterized by their dual nature, i.e., coexistence of yielded (fluid-like) and unyielded (solid-like) regions depending upon the existing stress levels vis-a-vis the fluid yield stress. This feature makes their homogenization and heating/cooling far more complicated than that for simple Newtonian fluids like air and water [1]. *In this work, the heat transfer characteristics for the steady flow of Bingham plastic fluids over a square cylinder confined in a plane slit have been explored numerically in terms of the isotherm contours, local Nusselt number and average Nusselt number for the following ranges of conditions: Bingham number, $0.01 \le Bn \le 100$, Reynolds number, $0.1 \le Re \le 40$, Prandtl number, $10 \le Pr \le 100$, and blockage ratio (ratio of side of cylinder to channel height), $\beta(l/H)$ = 0.2, 0.3, 0.4. Furthermore, the influence of the two different thermal boundary conditions i.e. constant heat flux condition (CHF) and constant wall temperature (CWT) prescribed on the surface of the square cylinder has also been delineated. The emphasis is laid on elucidating the effect of blockage ratio on the overall heat transfer as blockage ratio influences the temperature gradients and due to the extra dissipation at the confining walls. In Bingham plastic fluids, heat transfer occurs by two competing mechanisms i.e., by convection in fluid-like and by conduction in solid unyielded regions. Thus the overall heat transfer is determined in a complex manner by these two competing mechanisms. Finally, a simple predictive correlation is presented which captures the effect of Re, Bn, Pr and β on the average Nusselt number (in terms of the Colburn j-factor) for both CHF and CWT conditions.*

NOMENCLATURE

Bn	Bingham number $(=\tau_0 l/\mu_b U_0)$, dimensionless
C_D	drag coefficient, dimensionless
C_{DP}	pressure drag coefficient, dimensionless
C	specific heat of fluid, J/kg K
H	channel height, m
h	local heat transfer coefficient, W/m^2 K
j	Colburn factor for heat transfer, dimensionless
k	thermal conductivity of fluid, W/m K
l	side length of the square cylinder, m
L_u	upstream length, m
L_d	downstream length, m
m	growth rate parameter, dimensionless
n_s	unit vector normal to the cylinder surface, dimensionless
Nu_L	local Nusselt number, dimensionless
Nu	average Nusselt number, dimensionless
Pr	Prandtl number $(=C\mu_b/k)$, dimensionless
Pe	Peclet number, $(=Re\cdot Pr)$ dimensionless
q_s	constant heat flux on cylinder surface, W/m^2
Re	Reynolds number $(=\rho l U_0/\mu_b)$, dimensionless
S	surface area, m^2
T	temperature of fluid, K
T_w	cylinder wall temperature, K
T_0	temperature of the fluid at the inlet, K
U	Velocity vector, dimensionless
U_0	average inlet velocity, m/s
X,Y,Z	cartesian coordinates, dimensionless

Greek symbols

β	blockage ratio, dimensionless
$\dot{\gamma}$	rate of strain tensor, dimensionless
∇	del operator, m^{-1}
η	effective viscosity, dimensionless
μ_b	plastic viscosity, Pa s
ξ	temperature of fluid, dimensionless
ρ	fluid density, kg/m^3
τ	extra stress tensor, dimensionless
τ_0	yield stress, Pa

Abbreviations

CHF	constant heat flux
CWT	constant wall temperature

INTRODUCTION

Owing to the wide occurrence of viscoplastic fluids in various industrial applications, renewed interest has developed in studying their momentum and heat transfer characteristics past bluff bodies. An important class of viscoplastic fluids is constituted by the Bingham plastic fluid [1]. These fluids exhibit a threshold level of stress (yield stress) which must be overcome before a significant deformation occurs.

A cursory inspection of the literature shows that much research effort has been expended in studying the momentum/heat transfer characteristics for an unconfined circular, square and elliptical cylinder submerged in viscoplastic fluids [2-6]. Over the past decades there has been a spurt in research activity focussing on the effect of confinement or wall blockage on the flow phenomenon in non-Newtonian fluids. However, most of the available information which relates to the effect of confinement on the flow past a cylinder relates to power-law fluids [7,8]. There are very few studies [10,11] for this configuration in the viscoplastic medium. Mitsoulis [10] explored the effect of wall blockage on flow of Bingham plastic fluids past a circular cylinder in the creeping flow regime. While Mishra and Chhabra [11] studied the effect of confinement for a square cylinder in the steady flow regime $\left(Re \le 40\right)$. Both of these studies provide useful insights into the flow kinematics like morphology of the yielded/unyielded regions, streamlines, shear rate contours and the total and pressure drag coefficients. The present work is an extension to our recent study [11] and it deals with the effect of confinement on the forced convection heat transfer from a heated square cylinder immersed in a Bingham plastic fluid. The momentum and energy equations have been solved over a wide range of conditions as: Reynolds number, $0.1 \le Re \le 40$; Prandtl number, $10 \le Pr \le 100$; Bingham number $0.01 \le Bn \le 100$; blockage ratio, β = 0.2, 0.3 and 0.4. Over these range of conditions, the flow field is expected to be steady.

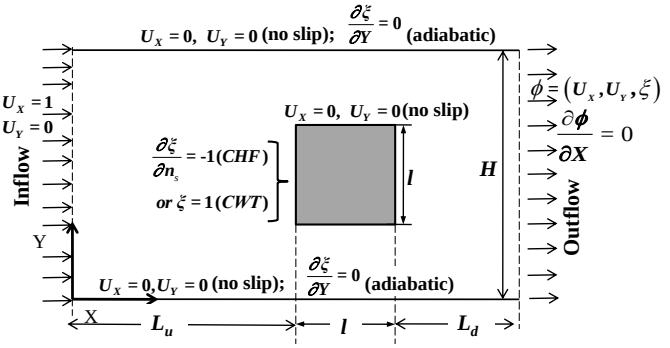


Figure 1. SCHEMATIC OF COMPUTATIONAL DOMAIN.

PROBLEM FORMULATION

A Bingham plastic fluid enters the channel with a uniform velocity, U_0 and at a constant temperature, T_0 at the inlet (shown in Figure 1). The surface of the square cylinder is maintained either at a constant heat flux (*CHF*) of q_s (cylinder dissipating heat at a constant rate) or it is maintained at a constant wall temperature (*CWT*) of T_w (where $T_w > T_0$) conditions. The inlet section is sufficiently long for the velocity profile to be fully developed before impinging on the square cylinder. A zero diffusion flux condition for all dependent variables (except pressure) is prescribed at the outlet and the usual no-slip condition is used on all solid boundaries. The cylinder and the channel walls are considered to be infinitely long in the Z-direction, hence the flow is approximated as two dimensional i.e. U_z = 0 and $\frac{\partial U_z}{\partial Z} = 0$. The blockage ratio, $\beta(l/H)$ is varied from 0.2 to 0.4. The flow is assumed to be steady, laminar, 2-D and incompressible and that the thermophysical properties of the fluid are constant with negligible viscous dissipation. Under these conditions the continuity, momentum and energy equations in their dimensionless forms are given as:

Continuity equation:

$$\nabla . \boldsymbol{V} = 0 \tag{1}$$

Momentum equation:

$$(\boldsymbol{V} \cdot \nabla)\, \boldsymbol{V} = -\nabla P + \frac{1}{Re} \nabla \cdot \tau \tag{2}$$

Energy equation:

$$(\boldsymbol{V} \cdot \nabla)\, \xi = \frac{1}{Re \cdot Pr} \nabla^2 \xi \tag{3}$$

The extra stress tensor, τ is given by the Bingham constitutive relation as [1]:

$$\tau = \left(1 + \frac{Bn}{|\dot{\gamma}|}\right)\dot{\gamma}\ , \quad \text{if} \quad |\tau| > Bn \tag{4a}$$

$$\dot{\gamma} = 0\ , \quad \text{if} \quad |\tau| \le Bn \tag{4b}$$

In aforementioned equations $\dot{\gamma}$, is the rate of deformation tensor and $|\dot{\gamma}|$ is the magnitude of the rate of deformation tensor given by:

$$|\dot{\gamma}| = \sqrt{tr\left(\dot{\gamma}^2\right)} \tag{5}$$

For an incompressible fluid, the extra stress tensor τ is related to the scalar viscosity function (η) as:

$$\tau = \eta \dot{\gamma}\ . \tag{6}$$

Since the Bingham constitutive relation (Eq. 4) is discontinuous and non-differentiable in nature, its direct implementation in a numerical scheme is rather difficult. One of the widely used regularization approach to circumvent this difficulty is the Papanastasiou's exponential method [9]. In this approach, the Bingham constitutive relation is regularized as:

$$\tau = \left(1 + \frac{Bn\left[1 - \exp\left(-m|\dot{\gamma}|\right)\right]}{|\dot{\gamma}|}\right)\dot{\gamma} \tag{7}$$

The scalar viscosity (η_P) is given as:

$$\eta_P = 1 + \frac{Bn\left[1 - \exp\left(-m|\dot{\gamma}|\right)\right]}{|\dot{\gamma}|} \tag{8}$$

Obviously, as the dimensionless regularization parameter $m \to \infty$, (Eq. 7) predicts the true Bingham plastic model. In the aforementioned equations, the length, velocity, pressure and the stress components have been reduced to their dimensionless forms using l, U_0, ρU_0^2 and $\mu_b(U_0 / l)$ respectively. Further

the temperature is non-dimensionalized as $\xi = (T - T_0)/(T_w - T_0)$ for the *CWT* condition and $\xi = (T - T_0)/(q_s l/k)$ for the *CHF* condition. In accordance with the dimensional considerations, the forced convection heat transfer characteristics in the present case are governed by four dimensionless numbers namely, Bingham number, Reynolds number, Prandtl number, and blockage ratio. The problem closure is obtained by specifying the following boundary conditions:

At the inlet:

$$U_X = 1, U_Y = 0, \xi = 0;$$

At the outlet:

$$\frac{\partial \phi}{\partial X} = 0, \text{where } \phi = U_X, U_Y \text{ and } \xi;$$

On the surface of the cylinder:

$$U_X = 0, U_Y = 0, \xi = 1(CWT) \text{ or } \frac{\partial \xi}{\partial n_s} = -1(CHF);$$

The numerical solution of the governing differential equations (Eqs. 1-3) together with the aforementioned boundary conditions maps the flow domain in terms of the primitive variables *u-v-p* and ξ. These, in turn, are post processed to evaluate the momentum and heat transfer characteristics. It is thus appropriate at this stage to define some of the gross parameters mentioned in the present study.

The local Nusselt number at a point on the surface of the cylinder is given as:

$$Nu_L = -\left.\frac{\partial \xi}{\partial n_s}\right|_{surface} \quad (CWT)$$

$$Nu_L = \left.\frac{1}{\xi}\right|_{surface} \quad (CHF)$$

The local Nusselt number is integrated over the surface of the cylinder to obtain the mean or average Nusselt number as:

$$Nu = \frac{hl}{k} = \frac{1}{S}\int_S Nu_L dS$$

The average value of the Nusselt number can be used to estimate the rate of heat transfer between an object and the surrounding fluid. Since the objective of the present study is to study the heat transfer characteristics, the fluid mechanical aspects such as the total and pressure drag coefficients are of little interest here and these have been used here only to ensure that the present results are free from the numerical artefacts and to select optimum values of the numerical parameters like domain size, grid and regularization parameter, *m*.

NUMERICAL SOLUTION SCHEME

The aforementioned non-linear continuity, momentum and energy equations have been solved using the finite-element based method COMSOL Multiphysics (Version 4.3a). A relative convergence criterion of 10^{-5} is used for the primitive variables (*u-v-p* and ξ). The von Mises criterion [1] with a relative tolerance level of 10^{-6} is used for approximating the yield surfaces. The numerical solution methodology used in the present work is similar to the one used in our recent study [11] hence it is described here in brief only. Following the strategy employed in our recent study [11], a series of numerical experiments were performed to optimize the domain (upstream and downstream length), grid and regularization parameter, *m*. Table 1 shows the effect of domain on average Nusselt number (Nu_{CHF} and Nu_{CWT}). Since these values varied slightly (within ±0.5%), hence suffice to say that L_u/l = 60 and L_d/l = 20 are optimum for the present study. Also Figure 2 shows the effect the velocity profile for a range of values of X/l = 20, 30, 40 and 50 for the extreme values of the Reynolds and Bingham numbers. It can be clearly observed that L_u/l = 60 is sufficient for the velocity profile to be fully developed. The present numerical values for average velocity in channel flow were compared with the expected analytical values of average velocity in Bingham plastic fluids by Chhabra and Richardson [12]. Furthermore, Table 2 shows the effect of three computational grids (*G1*, *G2* and *G3*) on the present results. Since the values of average Nusselt number (Nu_{CHF} and Nu_{CWT}) for grid *G2* and *G3* vary by <1%; hence grid *G2* is adequate for the present study. Although, large values of *m* are desirable to predict the true Bingham model however extremely large values of *m* can lead to numerical instabilities thereby further lending difficulty in solution convergence. Hence, an adequate value of the regularization parameter (*m*) is required. The specifics for *m* test are shown in Table 3. It can be clearly observed from Table 3 that the drag coefficients and average Nusselt number values change negligibly for $m \geq 10^5$ hence, $m = 10^5$ is adequate for the present study. In summary, L_u/l = 60 and L_d/l = 20, grid *G2* and regularization parameter, $m = 10^5$ have been found to be optimum for the present study to ensure that the present results are free from numerical artifacts.

Table 1. EFFECT OF UPSTREAM LENGTH, L_u/l AND DOWNSTREAM LENGTH, L_d/l AT *Re* = 0.1, *Pr* = 10 AND β = 0.2.

		Bn = 0.01		*Bn* = 100	
L_u/l	L_d/l	Nu_{CHF}	Nu_{CWT}	Nu_{CHF}	Nu_{CWT}
50	20	0.9121	0.5794	0.9158	0.5794
60	20	0.9120	0.5793	0.9159	0.5784
70	20	0.9120	0.5793	0.9159	0.5784
60	10	0.9121	0.5794	0.9158	0.5783
60	20	0.9120	0.5793	0.9159	0.5783
60	30	0.9120	0.5793	0.9159	0.5783

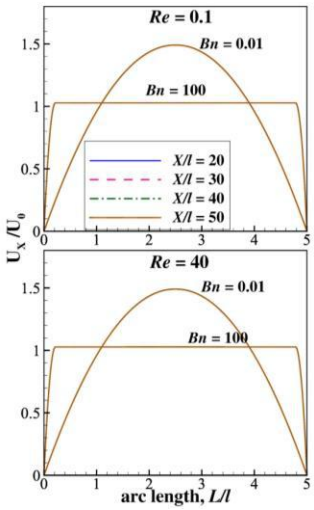


Figure 2. EFFECT OF UPSTREAM LENGTH, L_u/l = 60 ON THE VELOCITY PROFILE AT β = 0.2.

Table 2. GRID INDEPENDENCE TEST AT Re = 40, Pr = 100 AND β = 0.4.

		No of	Bn = 0.01		Bn = 100	
Grid	Np	elements	Nu_{CHF}	Nu_{CWT}	Nu_{CHF}	Nu_{CWT}
G1	400	78000	24.404	20.983	20.665	19.780
G2	480	89600	24.412	20.980	20.643	19.777
G3	560	101200	24.412	20.980	20.626	19.777

Np is the number of elements on cylinder surface

Table 3. INFLUENCE OF REGULARIZATION PARAMETER ON THE DRAG AND NUSSELT NUMBER AT Re = 40, Pr = 100 AND β = 0.4.

Bn = 10				
m	C_D	C_{DP}	Nu_{CHF}	Nu_{CWT}
10^4	16.529	13.692	21.824	20.678
10^5	16.529	13.687	21.338	20.674
10^6	16.529	13.687	21.338	20.674
Bn = 100				
10^4	91.932	77.503	20.967	20.668
10^5	92.031	76.692	20.908	20.665
10^6	92.031	76.588	20.665	20.665

Table 4. COMPARISON OF AVERAGE NUSSELT NUMBER IN NEWTONIAN FLUIDS FOR CONFINED SQUARE CYLINDER AT Pr = 100 AND β = 0.25.

	Dhiman et al. [8]		Present	
Re	Nu_{CHF}	Nu_{CWT}	Nu_{CHF}	Nu_{CWT}
10	9.7778	8.7293	9.7332	8.7461
20	13.333	11.492	13.321	11.429
40	18.556	15.407	18.512	15.264

RESULTS AND DISCUSSION

The aim of the present study is to elucidate the effect of confinement on the forced convection heat transfer from a heated square cylinder submerged in Bingham plastic fluid. In particular the effect of yield stress, Reynolds ($0.1 \le Re \le 40$), Bingham ($0.01 \le Bn \le 100$) and Prandtl numbers ($10 \le Pr \le 100$) on the heat transfer characteristics have been analysed in terms of isotherm contours, local Nusselt number and average Nusselt number for two different thermal boundary conditions, i.e., *CHF* and *CWT* for three values of the blockage ratio, β = 0.2, 0.3, 0.4. Initially an understanding of the temperature field is facilitated by examining the isotherm patterns for extreme values of the pertinent parameters Re, Pr and Bn for a fixed value of blockage ratio, β = 0.4 and then the level of complexity is built up to accommodate the influence of blockage ratio on heat transfer which is elaborately analysed here in terms of the local and average Nusselt number. However, before embarking upon the detailed discussion of the new results, it is desirable to establish the accuracy and reliability of the present results in the next section.

Validation of results

An extensive validation dealing with the fluid mechanical aspects for the case of an unconfined square cylinder in Bingham plastic fluid [4] and confined square cylinder in Newtonian fluids [8] has already been reported in our recent study [11], hence only additional comparisons are included here. The heat transfer characteristics (i.e., local Nusselt number values, Figure 3) for an unconfined square submerged in Bingham plastic fluid were compared with that of Nirmalkar et al. [4] and a good correspondence is obtained (within ±1-2%). Next, the present numerical results for a square cylinder placed in a channel were compared with that of Dhiman et al. [8] and a close correspondence was obtained (within ±1-2%), as seen in Table 4. Apart from the fact that the present study deals with the Bingham plastic fluid, otherwise on account of the geometric configuration and imposed thermal boundary conditions (on cylinder surface), this study is an extension of that of Dhiman et al. [8]. Hence the validation with this study lends additional credibility to the present results. Thus, it can be emphasized that an excellent agreement exists between the present numerical results and the available literature [4,8] hence the present numerical results are regarded to be reliable to within (±1-2%).

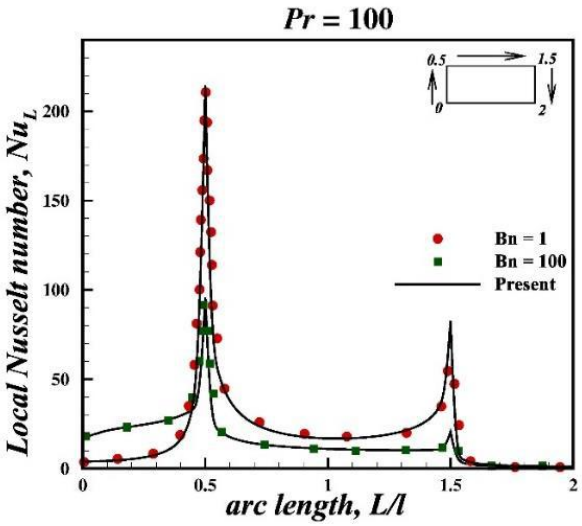


Figure 3. COMPARISON OF PRESENT RESULTS ON LOCAL NUSSELT NUMBER FOR AN UNCONFINED SQUARE CYLINDER IN BINGHAM PLASTIC FLUIDS WITH THAT OF NIRMALKAR ET AL. [4] AT Re = 40.

Isotherm contours

It is customary to visualize the temperature field in terms of the isotherm contours. Representative isotherm contours for both the *CHF* and *CWT* conditions are shown in Figures 4 and 5 for blockage ratio, β = 0.4 and extreme values of the Reynolds number, i.e., at Re = 0.1 and Re = 40 at Pr = 100. At Re = 0.1, it can be observed that isotherm contours are denser at the upstream side while they further expand out at the downstream side. This suggests that heat transfer is higher from the front surface compared to the rear surface. At Re = 0.1, no significant change is observed in the isotherm patterns as the value of Bingham number is increased progressively from Bn = 0.01 to Bn = 1 and further to Bn = 100. At Re = 40, notable distortion of isotherms is observed in the rear of the cylinder which is presumably due to the formation of a wake at the rear surface of the cylinder as the temperature field is influenced by the existing velocity field. However, as Bingham number increases, the yield stress increases and suppresses the propensity of wake formation and thus the distortion of isotherms decreases at higher Bingham number. For instance at Re = 40, irrespective of the boundary condition *CHF* or *CWT*, at Bn = 0.01 the distortion of isotherm contours is observed in the rear surface whereas no such distortion is observed at Bn = 100. Undoubtedly, in Bingham plastic fluids the heat transfer takes

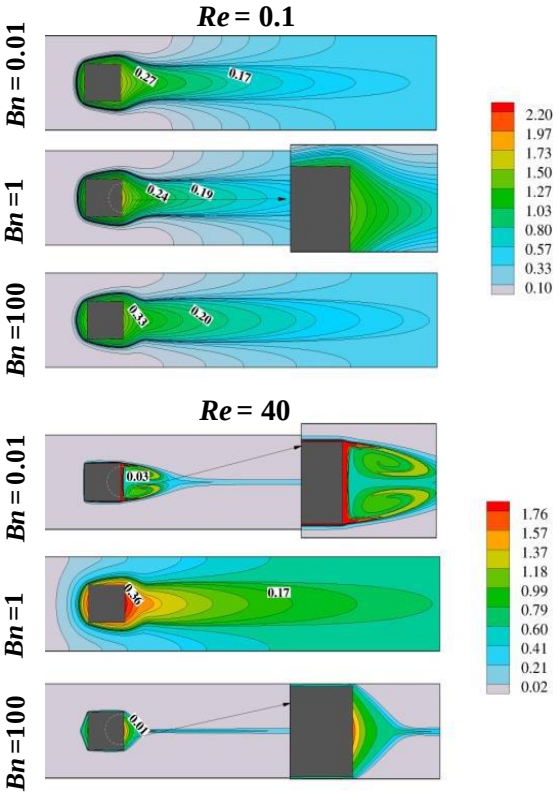


Figure 4. REPRESENTATIVE ISOTHERM PLOTS FOR CHF AT Pr = 100 AND β = 0.4

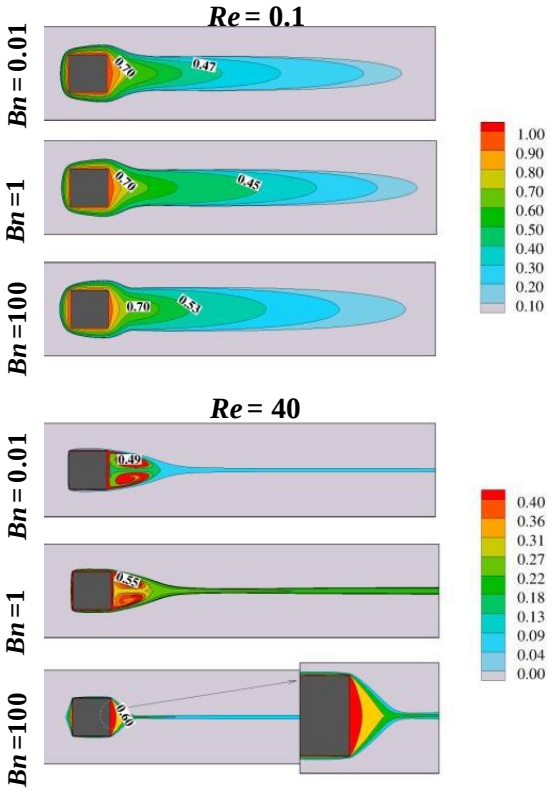


Figure 5. REPRESENTATIVE ISOTHERM PLOTS FOR CWT AT Pr = 100 AND β = 0.4.

place by two mechanisms, i.e., by conduction in solid unyielded regions and by convection in fluid-like regions. Thus, the complex isotherm patterns are determined by an intricate relationship between inertial, viscous and yield stress effects along with the varying contributions of conduction and convection effects. Since the isotherm patterns for other values of blockage ratio β = 0.2 and β = 0.3 were qualitatively similar to β = 0.4, these are not presented here.

Local Nusselt number

The value of the local Nusselt number is obtained by the corresponding temperature gradient normal to the surface of the heated cylinder. In the case of an isothermal cylinder (*CWT*), the temperature remains constant over the surface while in the case of isoflux cylinder (*CHF*), the temperature varies along the surface of the cylinder; however, in both cases, the temperature gradient varies from one point to another along the surface of the cylinder. Typical variation of the local Nusselt number over the surface of the heated square cylinder for both *CHF* and *CWT* conditions is shown in Figures 6 and 7 respectively for the blockage ratio, β = 0.4 and Reynolds number, Re = 40 and for the extreme values of Prandtl number, Pr = 10 and Pr = 100. It can be observed that the local Nusselt number exhibits positive dependence on the Bingham and Prandtl numbers. The increase in heat transfer with Prandtl number is due to the thinning of thermal boundary layers. Similarly, the positive dependence of the heat transfer on Bingham number can be ascribed to the fact that as the yielded regions diminish in size, steep temperature gradients arise in the vicinity of the heated cylinder which enhance the rate of heat transfer. Furthermore, it can be noted from Figures 6 and 7 that there is a sharp variation at the corners of the square cylinder which is due to the fact that the temperature gradient is indeterminate at these points. Strictly speaking, the local Nusselt number should exhibit a discontinuity at these singular points but as the computational mesh used in the present study is very fine such a discontinuity

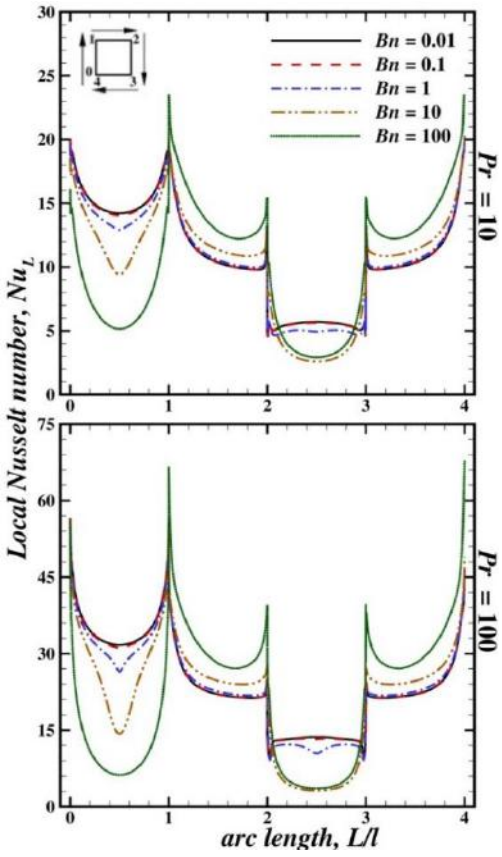


Figure 6. VARIATION OF THE LOCAL NUSSELT NUMBER ALONG THE SQUARE CYLINDER FOR CHF AT Re = 40 AND β = 0.4.

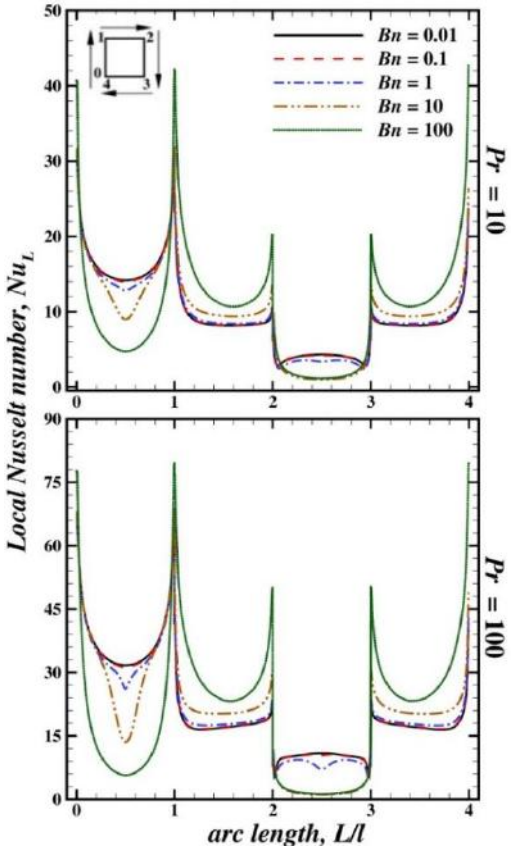


Figure 7. VARIATION OF THE LOCAL NUSSELT NUMBER ALONG THE SQUARE CYLINDER FOR CWT AT Re = 40 AND β = 0.4.

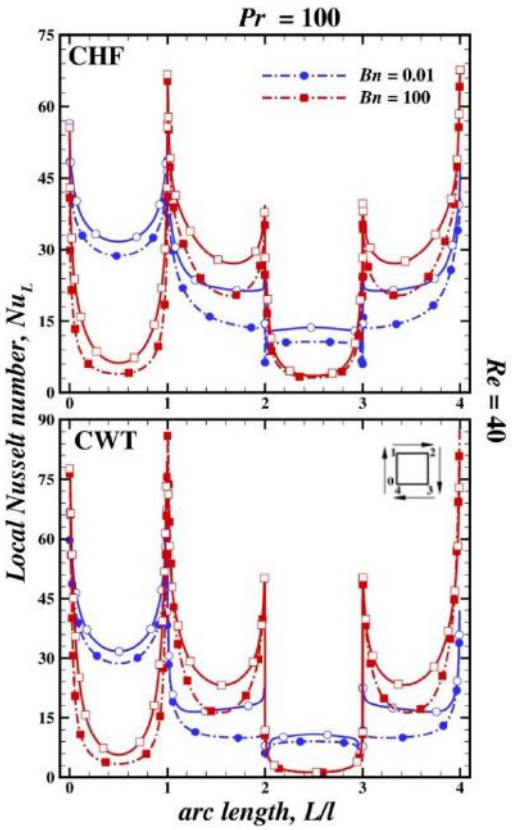


Figure 8. DEPENDENCE OF THE LOCAL NUSSELT NUMBER ON BLOCKAGE RATIO FOR β = 0.2 (FILLED SYMBOLS) AND β = 0.4 (UNFILLED SYMBOLS).

is not evident herein. Also, the heat transfer area is identically zero at these points hence the high values of the local Nusselt number at these points are of no practical significance. It can be noted from Figure 8 that the distribution of local Nusselt number along the surface of the square cylinder for β = 0.2 is qualitatively similar to β = 0.4 for both the thermal boundary conditions. However, it is worthwhile to mention here that the local Nusselt number also bears positive dependence on blockage ratio which is presumably due to the sharpening of the temperature gradients and extra dissipation at the confining walls.

Average Nusselt number

The overall rate of heat transfer from the surface of any heated bluff body is estimated by the value of surface averaged Nusselt number. Hence bearing in mind the pragmatic significance of the average Nusselt number in various processes, the present results endeavor to elucidate the dependence of the average Nusselt number on the pertinent parameters *Re*, *Bn*, *Pr* and β in this section. It can be clearly noted from Figures 9 and 10 that irrespective of the value of *Bn*, both *Re* and *Pr* have a positive influence on heat transfer which is presumably due to the thinning of the momentum and thermal boundary layers. As already mentioned earlier that heat transfer in Bingham plastic fluid occurs by two competing mechanism i.e., by convection in fluid-like (which enhances the rate of heat transfer) and by conduction in solid unyielded regions (which impedes the rate of heat transfer). The trends borne out by average Nusselt number plots shown in Figures 9 and 10 qualitatively support this assertion.

Furthermore, the influence of blockage ratio on average Nusselt number can easily be analysed from Figure 10. As seen here, the blockage ratio has positive influence on the overall heat transfer. It was observed that as the value of the blockage ratio increases from β = 0.2 to β = 0.4 the overall increment in the rate of heat transfer, i.e., average Nusselt number values was ~ 25% for *CHF* and ~ 27% for *CWT* condition. This can be explained here by stating that any modification in the extent of confinement alters the flow field which directly influences the temperature field and hence the rate of heat transfer. In fact an increase in the value of blockage ratio leads to narrowing of the annular region and enhances the levels of shearing at the confining walls. This in turn lowers the effective viscosity and

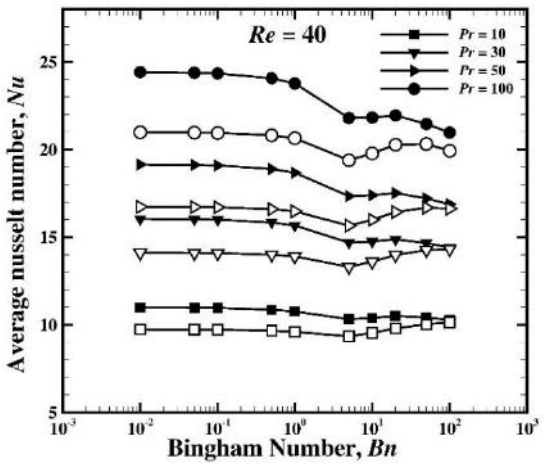


Figure 9. DEPENDENCE OF THE AVERAGE NUSSELT NUMBER ON PRANDTL NUMBER FOR *CHF* (FILLED SYMBOLS) AND *CWT* (UNFILLED SYMBOLS) CONDITIONS AT β = 0.4.

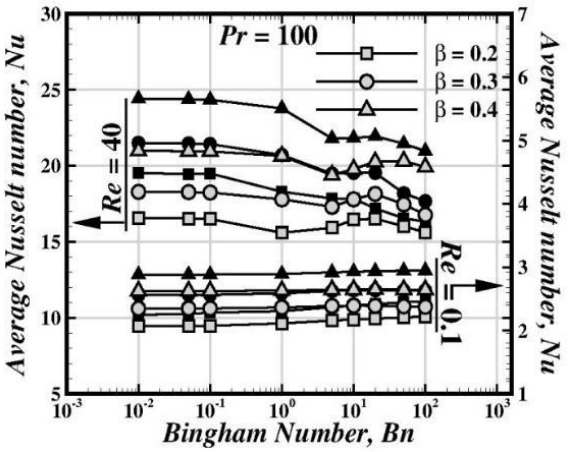


Figure 10. DEPENDENCE OF THE AVERAGE NUSSELT NUMBER ON BLOCKAGE RATIO FOR CHF (FILLED SYMBOLS) AND CWT (UNFILLED SYMBOLS) CONDITIONS.

hence facilitates heat transfer. Since this enhancement occurs simultaneously with the sharpening of the temperature gradients at the confining walls, the overall rate of heat transfer is further enhanced. Interestingly, these effects are further accentuated in Bingham plastic fluids owing to their complex rheological nature. Also, as shown in Figure 10, the rate of heat transfer is higher in the *CHF* condition as compared to that for the *CWT* condition.

Finally, in accordance with the dimensional considerations, the functional dependence of average Nusselt number on pertinent parameters *Re*, *Bn*, *Pr* and β can be expressed as:

$$Nu = f\left(Re, Pr, Bn, \beta\right)$$

Naturally this approach leads to a family of curves depending upon the corresponding values of *Re*, *Pr*, *Bn* and β. Following the strategy employed in our recent studies [4,11], the role of *Bn* is eliminated from the set of parameters by using the modified definitions of the Reynolds $\left[Re^{*} = Re/\left(1+Bn\right)\right]$ and Prandtl numbers $\left[Pr^{*} = Pr\left(1+Bn\right)\right]$. However, this format of presentation still leads to a family of curves. As such in order to consolidate the present numerical results for a wide range of Prandtl numbers into one single curve, the so called Colburn *j*-factor, defined below, has been used here.

$$j = \frac{Nu}{Re^{*} \times Pr^{*1/3}} \tag{9}$$

The results plotted in this fashion are shown in Figure 11. Finally the present numerical results have been correlated as follows:

$$j_{CHF} = 1.76\frac{\left(\beta\right)^{0.28}}{\left(Re^{*}\right)^{2/3}} \tag{10}$$

$$j_{CWT} = 1.63\frac{\left(\beta\right)^{0.23}}{\left(Re^{*}\right)^{2/3}} \tag{11}$$

Eq.10 and Eq.11 reproduce ~ 550 individual data points for both the *CHF* and *CWT* conditions respectively. The resulting mean and maximum deviations are found to be within 5% and 29% for *CHF* and 9% and 29% for *CWT* conditions respectively. For only 5 data points in each case (*CHF* or *CWT*) the deviation between the present and predicted value was found to be greater than 25%. These deviations correspond to lowest values of Reynolds and Prandtl numbers i.e. *Pe* ~ 1 which is indeed the conduction dominated regime. At the outset, it can be stated that these correlations (Eqs. 10 and 11) further support the qualitative trends borne out by the average Nusselt number (Figures 9 and 10) and colburn *j*-factor (Figure 11) on blockage ratio, thermal boundary conditions and other dimensionless parameters *Re*, *Pr* and *Bn* .

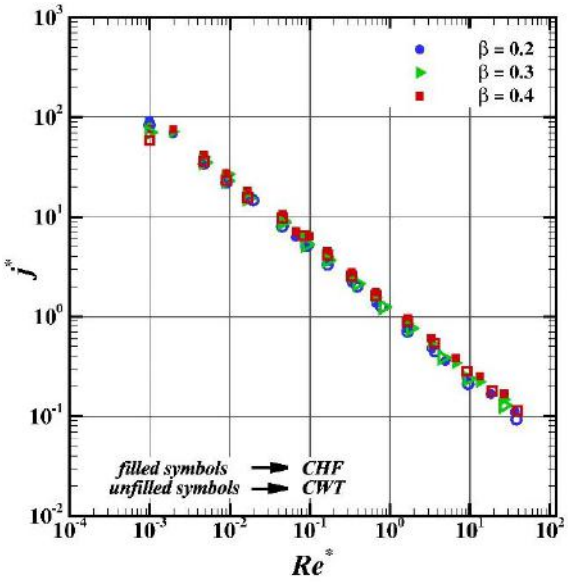


Figure 11. VARIATION OF *j*-FACTOR WITH MODIFIED REYNOLDS NUMBER (*Re**).

CONCLUSIONS

The effect of confinement on the forced convection heat transfer from a heated square cylinder submerged in Bingham plastic fluid are studied over wide ranges of conditions**:** $0.01 \le Bn \le 100$, $0.1 \le Re \le 40$, $10 \le Pr \le 100$, and $0.2 \le \beta \le 0.4$. In order to delineate the significance of these dimensionless parameters, extensive numerical results on isotherm contours, local and average Nusselt number have been discussed in detail. In Bingham plastic fluids, as Reynolds number increases, the size of the fluid-like cavity (where convection occurs) expands which enhances the rate of heat transfer whereas with the increasing Bingham number the size of solid-like unyielded regions (where conduction occurs) expand which impedes the rate of heat transfer. Thus the overall heat transfer is modulated by these two competing mechanisms. It was noted that for both thermal boundary conditions (*CHF* or *CWT*) the average Nusselt number bears a positive dependence on *Re*, *Pr* and β. However, it was found that under most conditions the rate of heat transfer was higher in *CHF* conditions than that in *CWT* condition. Also, an increase in the extent of confinement or blockage ratio enhances the overall rate of heat transfer. Finally, the present numerical results on the average Nusselt number have been correlated in terms of the Colburn *j*-factor using the modified Reynolds and Prandtl numbers.

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